Magnetic Resonance Imaging
361.2.6501
Final Project

Submission Date: 10.08.2026

1 Introduction

For the final project, you will choose one of three challenges. Each challenge focuses on a different type of MRI processing task: Brain-Age prediction, tumor segmentation, and K-space reconstruction. For your selected task, you are required to compare two models: the first should be a baseline model using a simple classical heuristic approach. The second must be your own original model or a meaningful variation of the baseline model. Full details on the report structure are provided below.

Note: We are aware that you are working with limited computational resources and limited time, so we do not expect state-of-the-art results. The grade will emphasize correct methodology, a meaningful comparison, reproducibility, and thoughtful analysis. Very poor performance may reduce the grade only when it reflects a flawed implementation, an inappropriate experiment, or insufficient analysis.

2 Datasets and HPC Servers

For this assignment, you will get access to the HPC servers of the university. These servers provide the computational power required for training deep neural networks.
Detailed instructions on how to work on the servers appear on the Moodle course page.
In addition, the servers will contain two datasets you can use for your projects:

1. **Brain Age/Reconstruction:** This dataset contains NumPy files containing 3D scans. The details of the scans, such as age, sex, etc., are specified in CSV files in the directory. There are three CSV files, divided into train, validation, and test samples, but these splits are not strict.

2. **BraTS dataset**: This dataset contains MRI scans and brain tumor segmentations from the BraTS 2020 challenge. Each case includes four MRI modalities: T1, T1ce, T2, and FLAIR. More about the dataset can be read on the Kaggle page and the official BraTS 2020 data page [1].

Note: If you prefer, you are allowed to use only the central slices of scans for the task. If you use this option, apply the same slice-selection rule to all methods and data splits, and describe the rule clearly in the report.

3 Project Descriptions

Choose one of the following projects. Each challenge focuses on a different MRI processing task and is evaluated using specific performance metrics. The references cited in the project descriptions and listed at the end of the document are provided as optional background reading only; you are not required to follow or implement the algorithms described in these papers.

3.1 Brain Age Prediction

In computational neuroscience, there has been an increased interest in developing algorithms that leverage brain imaging data to provide estimates of 'brain age' for an individual. Importantly, the discordance between brain age and chronological age (referred to as 'brain age gap') can capture accelerated aging due to adverse health conditions and therefore, can reflect increased vulnerability to neurological disease or cognitive impairments. As optional background, deep-learning brain-age models can also be analyzed to identify age-related regional morphological markers [3].

Task: Your model should take as input the MRI scan and output the estimated age of the patient.

Baseline example: A simple baseline may predict the mean age in the training set, or use linear regression on simple image-derived features such as intensity statistics or coarse morphology-inspired measurements.

Metrics for Evaluation: Create a scatter plot where each sample appears as a dot. The X-axis represents the true age of the patient. The Y-axis represents the predicted age of the patient. A red diagonal line represents perfect prediction. The title should say "Baseline/Our Model. MAE: _ Years, Pearson r: _". MAE stands for mean absolute error. Also report the Pearson correlation coefficient between the true and predicted ages.

3.2 MRI Restoration from Subsampled k-space

A main bottleneck of MRI is scan time: acquiring a full k-space volume can take several minutes, which reduces patient comfort and increases vulnerability to motion artifacts, and is especially limiting for time-sensitive imaging such as fetal, cardiac, or functional MRI. One way to accelerate acquisition is to sample only a fraction of the k-space and computationally restore the missing information, rather than acquiring it directly. However, naively skipping k-space samples introduces aliasing artifacts in the reconstructed image, since the undersampling violates the Nyquist criterion. Restoration algorithms aim to recover an anatomically faithful image from such partial, corrupted acquisitions, effectively decoupling scan time from image quality. For optional background reading on deep-learning-based MRI reconstruction, see [2].

Task: Given a fully sampled MRI slice, simulate an accelerated acquisition by subsampling its k-space using a 1D random variable-density mask: rows are sampled from a normal distribution centered on the middle of k-space (so low frequencies, which carry most of the image energy, are preserved, while high frequencies are sparsely sampled), keeping separately $20\%, 30\%$ and 50% of the original rows. Sample rows without replacement. Zero-fill the unsampled rows and apply an inverse Fourier transform to obtain the corrupted input image. Your model should take this zero-filled, aliased image as input and output a restored image that approximates the original, fully sampled scan.

Note: Use a random seed for reproducibility of the experiment.

Metrics for Evaluation: For each sampling ratio (20%, 30%, 50%) and each test sample, calculate the Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index (SSIM) between the restored image and the original, fully sampled image. For complex-valued images, compute PSNR and SSIM on the real and imaginary components separately. Normalize images consistently before computing these metrics. Report the mean and standard deviation of both metrics across the test set in a table, with one row per sampling ratio for both the baseline and your model. In addition, present two plots (one for PSNR, one for SSIM) with sampling ratio on the X-axis and the metric value on the Y-axis. Each plot should contain two lines: one for the baseline and one for your model, with shaded regions or error bars representing the standard deviation. In addition, include scatter plots of sample-wise baseline versus your model performance for PSNR and SSIM (for each sampling ratio, or with sampling ratios clearly marked), and report the Pearson correlation coefficient (r) on each scatter plot.

3.3 Brain Tumor Segmentation

Delineation of brain tumors from MRI scans is a foundational task that underlies many downstream analyses, including volumetric studies, disease progression tracking, and surgical planning. Automated segmentation algorithms aim to replace or assist the manual annotation process performed by radiologists, while maintaining high anatomical fidelity.

Task: Your model should take the four MRI modalities as input and output a segmentation mask. For this project, use binary whole-tumor segmentation, i.e., tumor versus background. If you choose to perform multi-class segmentation instead, clearly state this and report the metrics accordingly.

Metrics for Evaluation: For each test sample, calculate the Dice similarity coefficient (Dice) and Intersection over Union (IoU) between the predicted segmentation mask and the ground truth annotation. Dice is defined as $\text{Dice} = \frac{2|P \cap G|}{|P|+|G|}$, and IoU is defined as $\text{IoU} = \frac{|P \cap G|}{|P \cup G|}$, where P is the predicted mask and G is the ground truth mask. Report the mean and standard deviation of both metrics across the test set in a table, with one row for the baseline and one row for your model. In addition, present two box plots (one for Dice, one for IoU) showing the distribution of scores across the test set, with one box for the baseline and one box for your model. Also include scatter plots of sample-wise baseline versus your model performance for Dice and IoU, and report the Pearson correlation coefficient (r) on each scatter plot.

4 Evaluation Rubric

The project will be evaluated according to the following components. In all parts of the project, students should connect their choices and explanations to MRI concepts taught in the course, such as k-space sampling, image contrast, MRI modalities, artifacts, anatomical structures, and appropriate evaluation metrics.

Component	Points
MRI motivation and understanding of the selected task	10
Data description and MRI-specific preprocessing, including modalities, slice selection, normalization, or k-space masking where relevant	15
Baseline model grounded in MRI, anatomical, or physical reasoning	15
Proposed model or meaningful variation, including clear implementation details	20
Quantitative evaluation and visual presentation of results using the required MRI-relevant metrics	20
Interpretation of results in MRI terms, including artifacts, anatomical plausibility, limitations, and failure cases	10
Code quality, reproducibility, and README	10
Total	100

5 The Report

5.1 Title

The final report should contain:

1. Course name and number.
2. Final Project.
3. Names and student IDs of the submitters.
4. A link to a GitHub repository containing your code, along with a README or Markdown file that briefly presents the project. Do not upload the datasets or medical image files to GitHub. (The link can also be shared as part of a LinkedIn post or as a project on your CV!)

5.2 Abstract

Include a one-paragraph summary:

1. The challenge you chose.
2. The two models you used, with a brief description of each.
3. A summary of your conclusions.

5.3 Introduction, Objective, and Data Description

1. Clearly state the challenge.
2. Describe the dataset, including exploratory data analysis and visualizations.
3. Briefly explain the evaluation metrics used.

5.4 Baseline Model

For the task you chose, implement a classical, non-deep-learning baseline model, using anatomical or physical knowledge gained during the course. A few examples: measuring the size of age-indicative brain structures for age prediction; interpolation-based methods for MRI restoration; and classical image processing techniques for segmentation. Provide a short written explanation of your reasoning, along with a description of the model and a simple diagram of the pipeline.

5.5 Your Model

In this part you can choose between two options:

1. **Advanced classical algorithm:** You can use a more advanced variation of the baseline model, or alternatively use a different classical algorithm. Provide a short written explanation of your reasoning, along with a description of the model and a simple diagram of the pipeline.

2. **Deep learning model:** You can use deep learning for the task. If you choose this route, provide a description of the data splits you used, pre-processing, architecture, and loss function. Provide a simple diagram of the pipeline.

5.6 Results

1. Compare the performance of both models (both quantitative and qualitative) with the metrics described in the "Project Descriptions" section. Clearly state whether your variation outperformed the baseline model, and explain the reasons behind the outcome.

2. Show four examples of images. For each example, include the input image, the baseline output, your model output, the original image or ground-truth annotation, and a short explanation:

   - One where both models performed well.
   - One where both models performed poorly.
   - One where the baseline model performed well and your model did not (if none, please state).
   - One where your model performed well and the baseline model did not (if none, please state).

5.7 Conclusions and Summary

Summarize key findings and overall conclusions.

6 Submission Instructions

- Work in pairs; only one submission per team is required.
- Submit the report as a PDF file.

Document Instructions

- Reports may be written in English or Hebrew.
- Use the section titles provided in this document.
- All images, figures, and tables should be explained and clearly visible.
- Ensure the document is organized and easy to read.

Code Instructions

- Upload your code to a public GitHub repository. Do not upload the datasets or medical image files.
- Include a clear and concise README file with instructions on how to run the code.
- Ensure the code is modular, well-organized, and reusable.
- Begin each code file with the full names and student IDs of all team members.
- Use meaningful variable and function names, and include comments where necessary.
- Use only relative paths for file access—avoid hardcoded absolute paths.
- The code must run from start to finish without errors.

7 Background References

The following references are provided for background only. Students may consult them for context, but are not required to follow the algorithms or implementation choices described in these papers.

References

[1] Center for Biomedical Image Computing & Analytics (CBICA), University of Pennsylvania. *Multimodal Brain Tumor Segmentation Challenge 2020: Data*. `https://www.med.upenn.edu/cbica/brats2020/data.html`

[2] R. Shaul, I. David, O. Shitrit, and T. Riklin Raviv. *Subsampled brain MRI reconstruction by generative adversarial neural networks*. Medical Image Analysis, 65:101747, 2020. `https://doi.org/10.1016/j.media.2020.101747`

[3] G. Levakov, G. Rosenthal, I. Shelef, T. Rinklin Raviv, and G. Avidan. *From a deep learning model back to the brain—Identifying regional predictors and their relation to aging*. Human Brain Mapping, 41(12):3235–3252, 2020. `https://doi.org/10.1002/hbm.25011`